

System Administration and Management

Active Directory Domain Services

CODE: ET 8413

FACILITATOR: FRANK

EMAIL: frankibrahim25@gmail.com

Domain controller(DC)

- ❑ A Domain is a logical grouping of **users, systems, applications, and resources** that are managed under a single administrative structure.
- ❑ This structure allows for centralized management of security policies and user access rights.
- ❑ Domains are primarily used in **Microsoft Windows Active Directory** environments to centralize and streamline the management of resources, security, and authentication.

- ❑ A **domain controller** is a server running the Active Directory Domain Services (AD DS) role.
- ❑ A Domain Controller (DC) allows the creation of **logical containers**.
- ❑ These containers consist of users, computers and groups.
- ❑ When you install Windows Server on a computer, you can choose to configure a specific **server role** for that computer.

Active Directory Domain Services (AD DS)

- ❑ A directory is a hierarchical structure that stores information about objects on the network.
- ❑ A directory service, provides the methods for **storing directory data** and making this data **available** to network users and administrators.
- ❑ Active Directory (AD) is a directory service developed by Microsoft, primarily used for managing **permissions** and **access** to network resources within Windows domain networks.

- ❑ Active directory provide an authentication and authorization systems to network resources.
- ❑ The information stored in Active Directory is referred to **Active Directory objects** and makes this information easy for administrators and users to find and use.
- ❑ **Active Directory Objects:** These are entities within AD that represent users, groups, computers, printers, and other resources. Each object has attributes that define its characteristics and permissions.

- ❑ Inside the domain objects are classified into organizational units(OUs).
- ❑ **Organizational Units (OUs):** OUs are container objects within Active Directory that allow administrators to group related objects together.
- ❑ They provide a way to organize users, computers, and other resources based on common characteristics such as department, location, or function.

Active Directory Forest

- ❑ An Active Directory forest is the highest level of organization within the Active Directory (AD) structure, serving as a container for one or more domains that share a common schema and configuration.
- ❑ It represents a security boundary and provides a framework for managing network resources across multiple domains.

- ❑ A forest can contain multiple domains, each of which can have its own organizational units (OUs), user accounts, and resources.
- ❑ All domains within a forest share a common schema, which defines the types of objects that can exist in the directory and their attributes.
- ❑ This ensures consistency in how data is structured and accessed across the domains.
- ❑ The forest acts as a security boundary, meaning that objects in different forests cannot interact unless trust relationships are established.

Creating a new Active Directory Forest

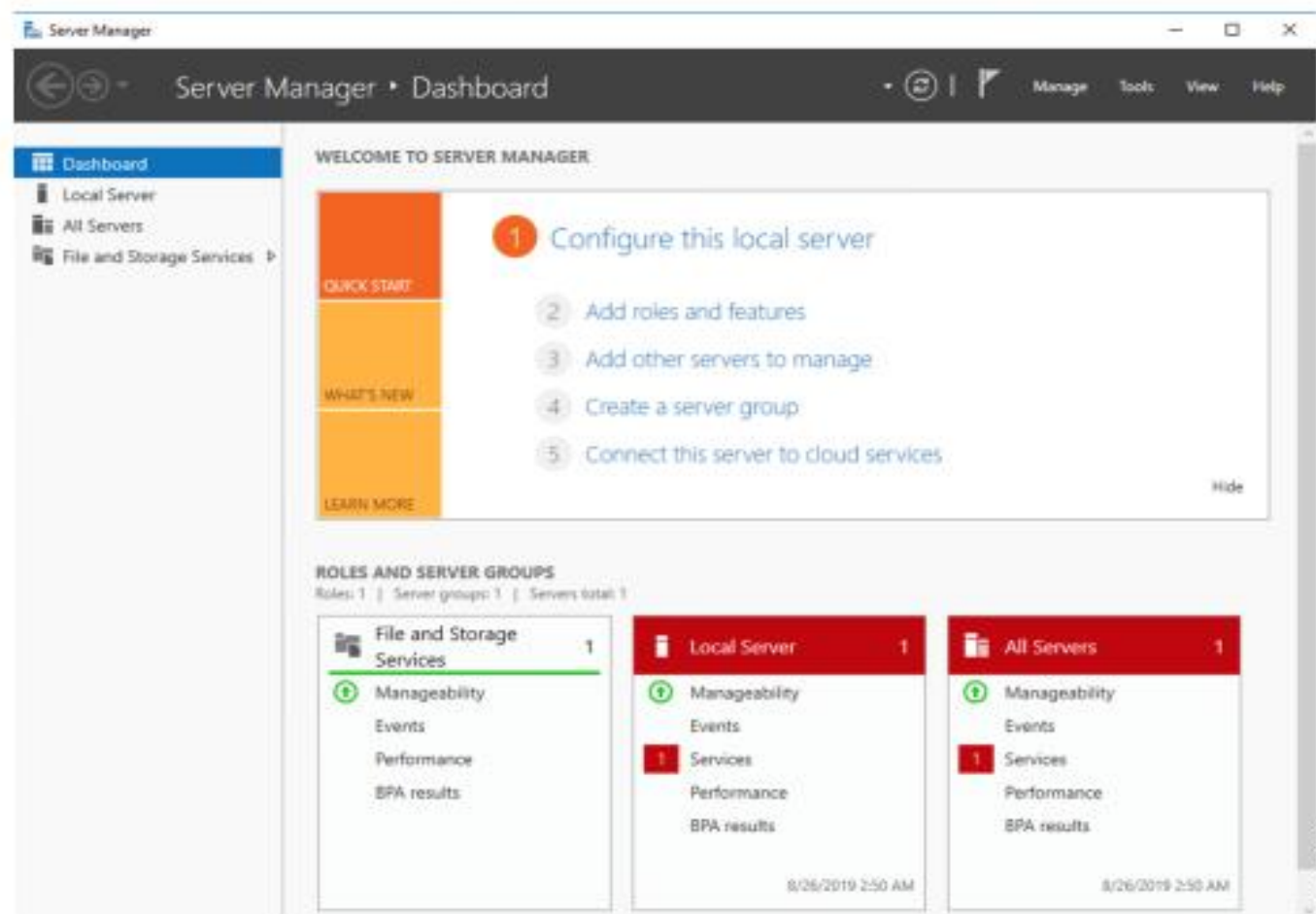
- ❑ Install Active Directory Domain Services (AD DS)
 - Open Server Manager.
 - Click on Add Roles and Features.
 - Follow the wizard to install the Active Directory Domain Services role.
- ❑ Promote Server to a Domain Controller:
 - Return to Server Manager and click on the notification.
 - Select Promote this server to a domain controller.
 - Choose Add a new forest and enter your desired root domain name (e.g., yourdomain.com).

Creating a new Active Directory Forest

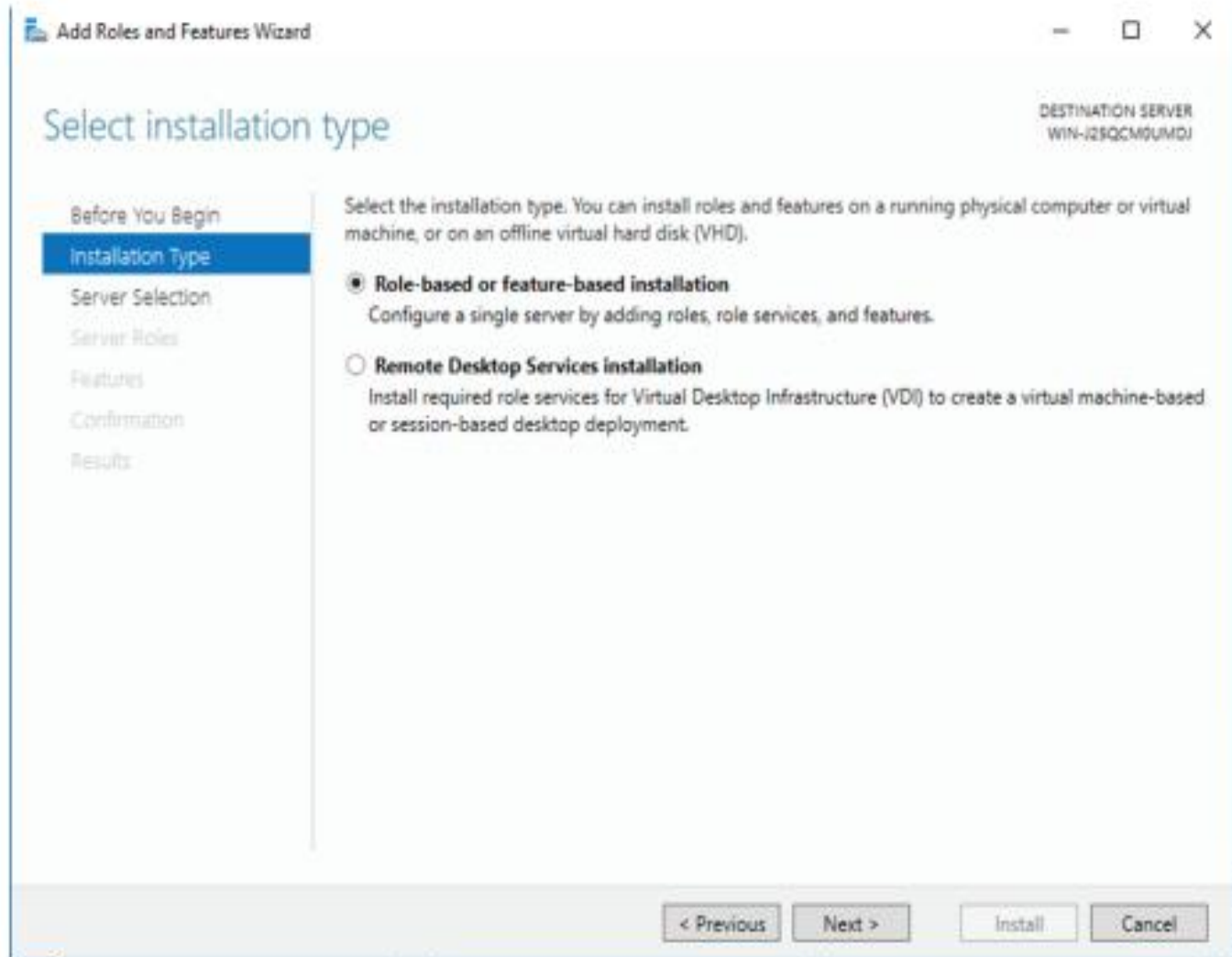
- ❑ Configure Forest and Domain Settings:
 - Set the forest and domain functional levels (choose the highest level supported by your environment).
 - Specify the Directory Services Restore Mode (DSRM) password.
 - Configure additional options as necessary, such as DNS settings.
- ❑ Review your configurations and click Install. The server will reboot upon completion, becoming the first domain controller and root of your new AD forest.

Install and configure Active Directory

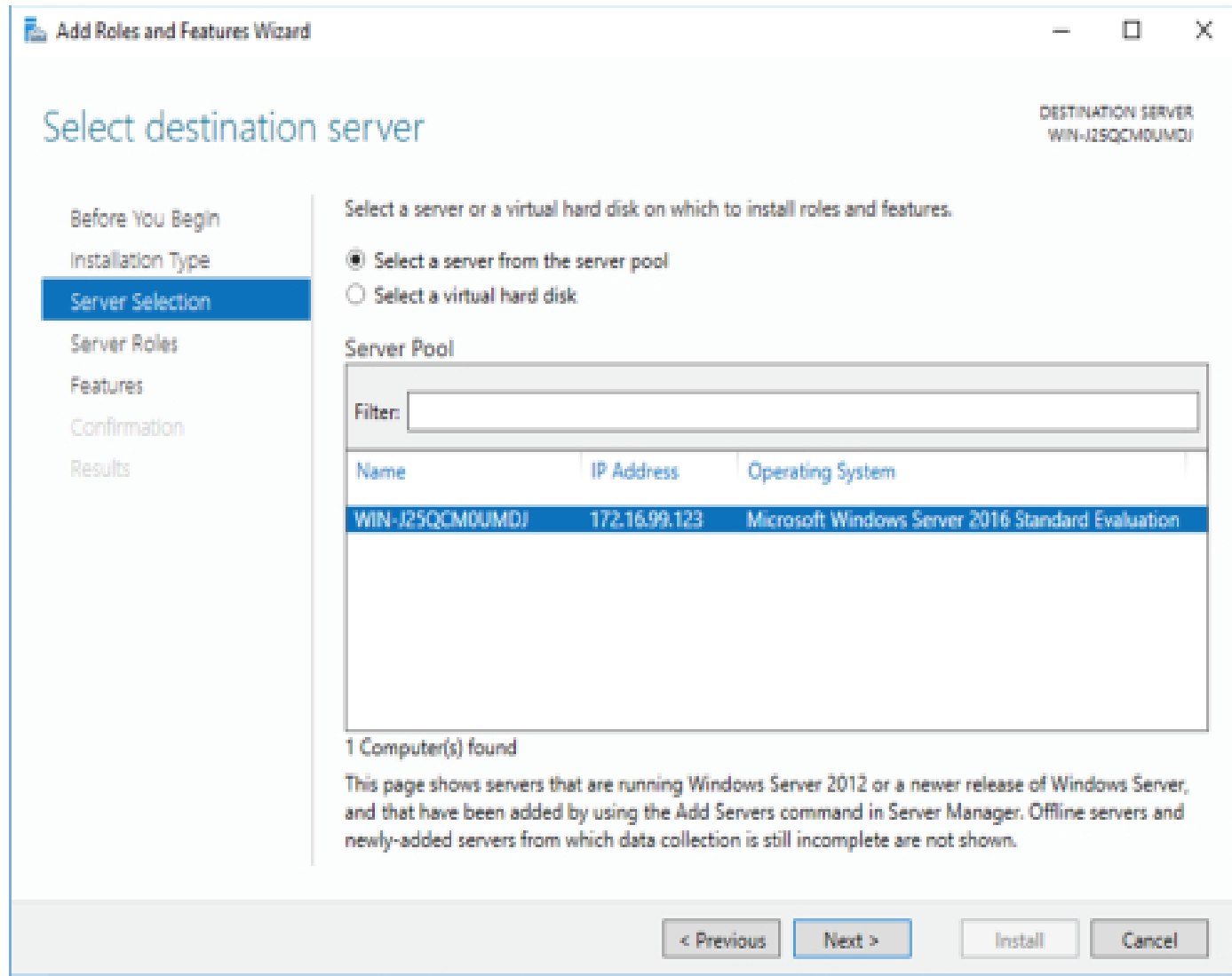
- ❑ Navigate to the Windows Server Manager.
- ❑ Click Add Roles and Features.



❑ It will open Add Roles and Features wizard. Click Next



Select the server from the server pool and click Next



The screenshot shows the 'Add Roles and Features Wizard' window. The title bar reads 'Add Roles and Features Wizard'. The main heading is 'Select destination server'. In the top right corner, it says 'DESTINATION SERVER: WIN-I25QCM0UMDI'. On the left, a navigation pane lists steps: 'Before You Begin', 'Installation Type', 'Server Selection' (highlighted), 'Server Roles', 'Features', 'Confirmation', and 'Results'. The main area contains the instruction 'Select a server or a virtual hard disk on which to install roles and features.' with two radio buttons: 'Select a server from the server pool' (selected) and 'Select a virtual hard disk'. Below this is a 'Server Pool' section with a 'Filter:' text box and a table. The table has three columns: 'Name', 'IP Address', and 'Operating System'. It contains one row: 'WIN-I25QCM0UMDI', '172.16.99.123', and 'Microsoft Windows Server 2016 Standard Evaluation'. Below the table, it says '1 Computer(s) found'. A descriptive paragraph follows: 'This page shows servers that are running Windows Server 2012 or a newer release of Windows Server, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.' At the bottom, there are four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results

Select a server or a virtual hard disk on which to install roles and features.

☒ Select a server from the server pool
☐ Select a virtual hard disk

Server Pool

Filter:

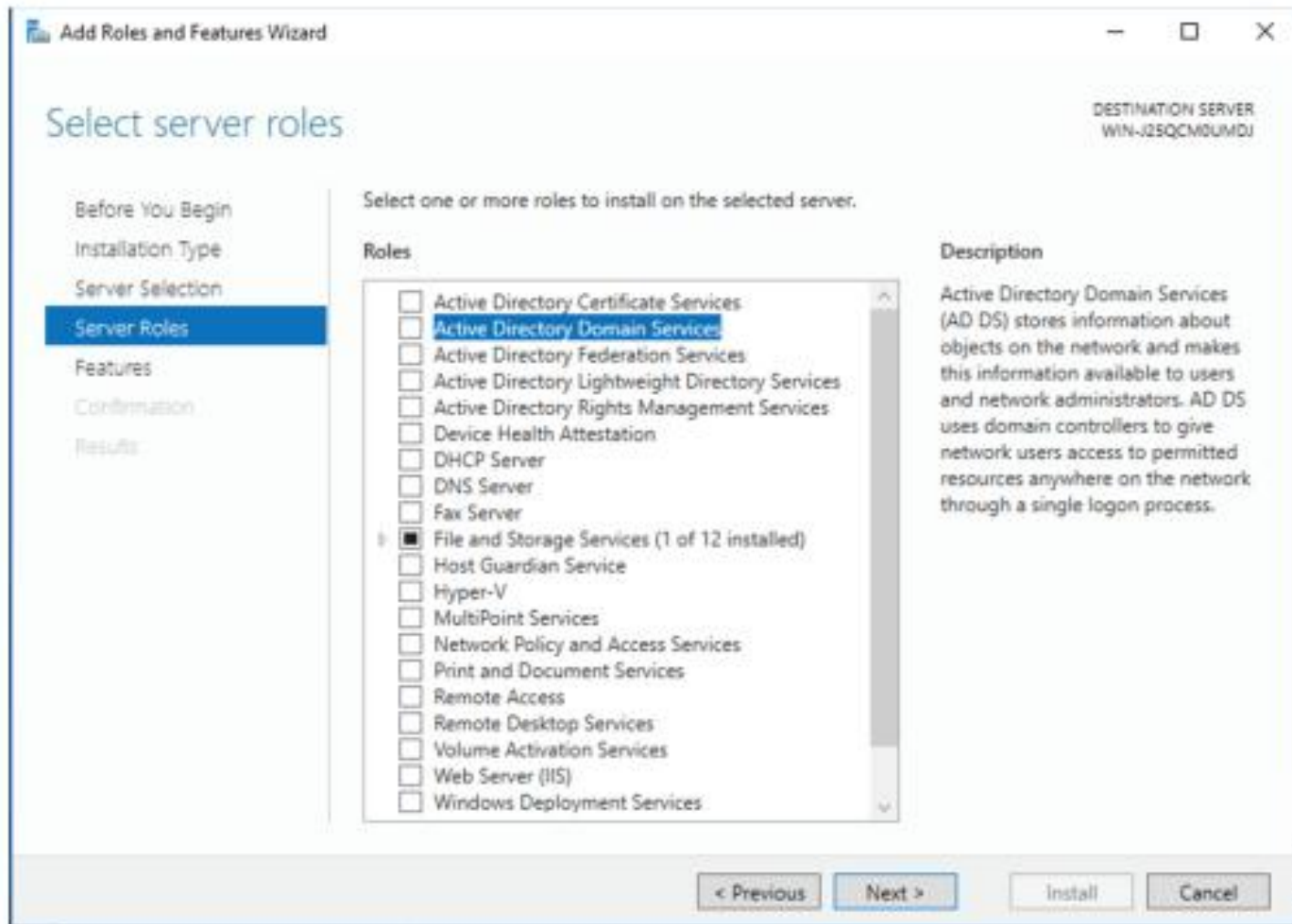
Name	IP Address	Operating System
WIN-I25QCM0UMDI	172.16.99.123	Microsoft Windows Server 2016 Standard Evaluation

1 Computer(s) found

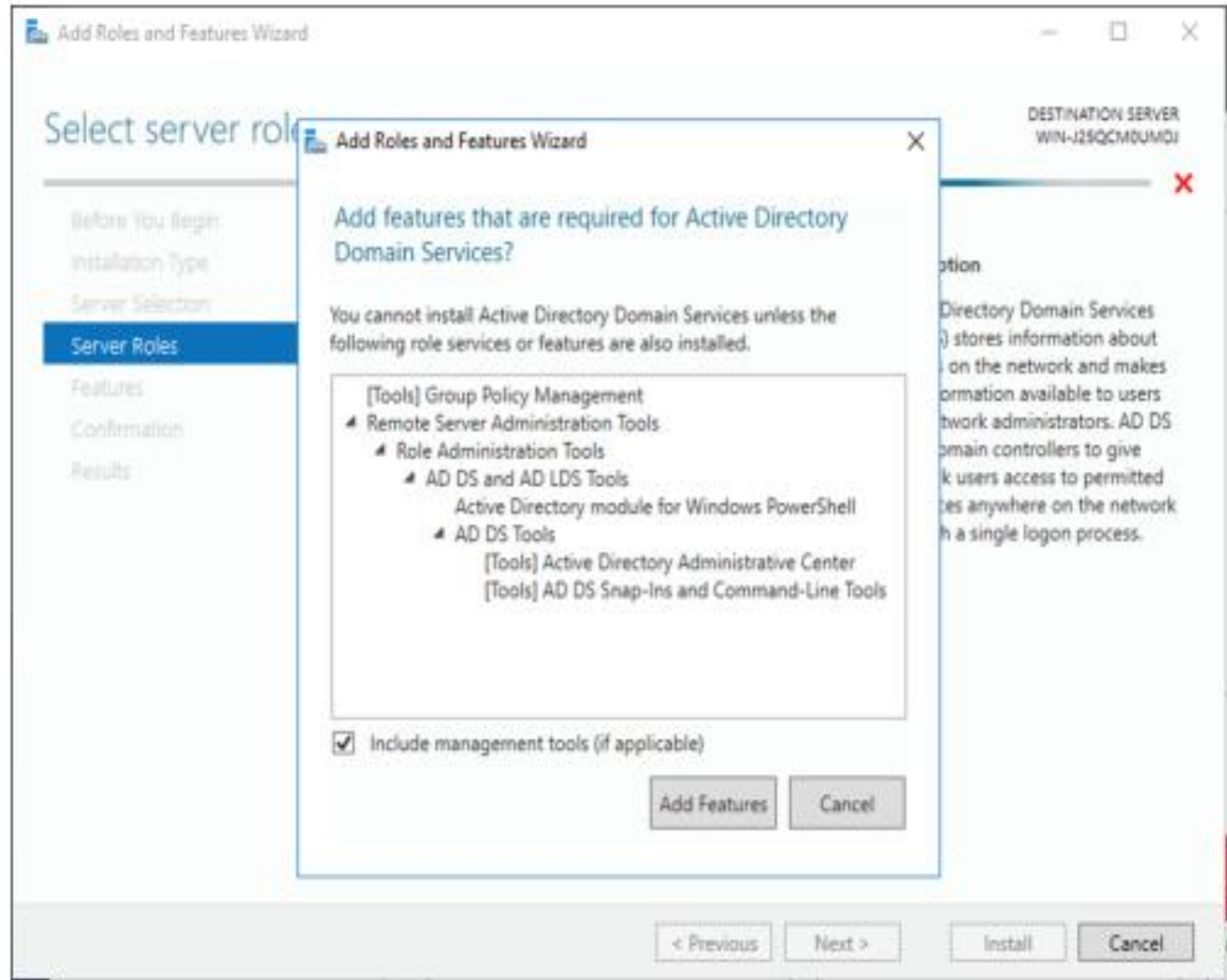
This page shows servers that are running Windows Server 2012 or a newer release of Windows Server, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.

< Previous Next > Install Cancel

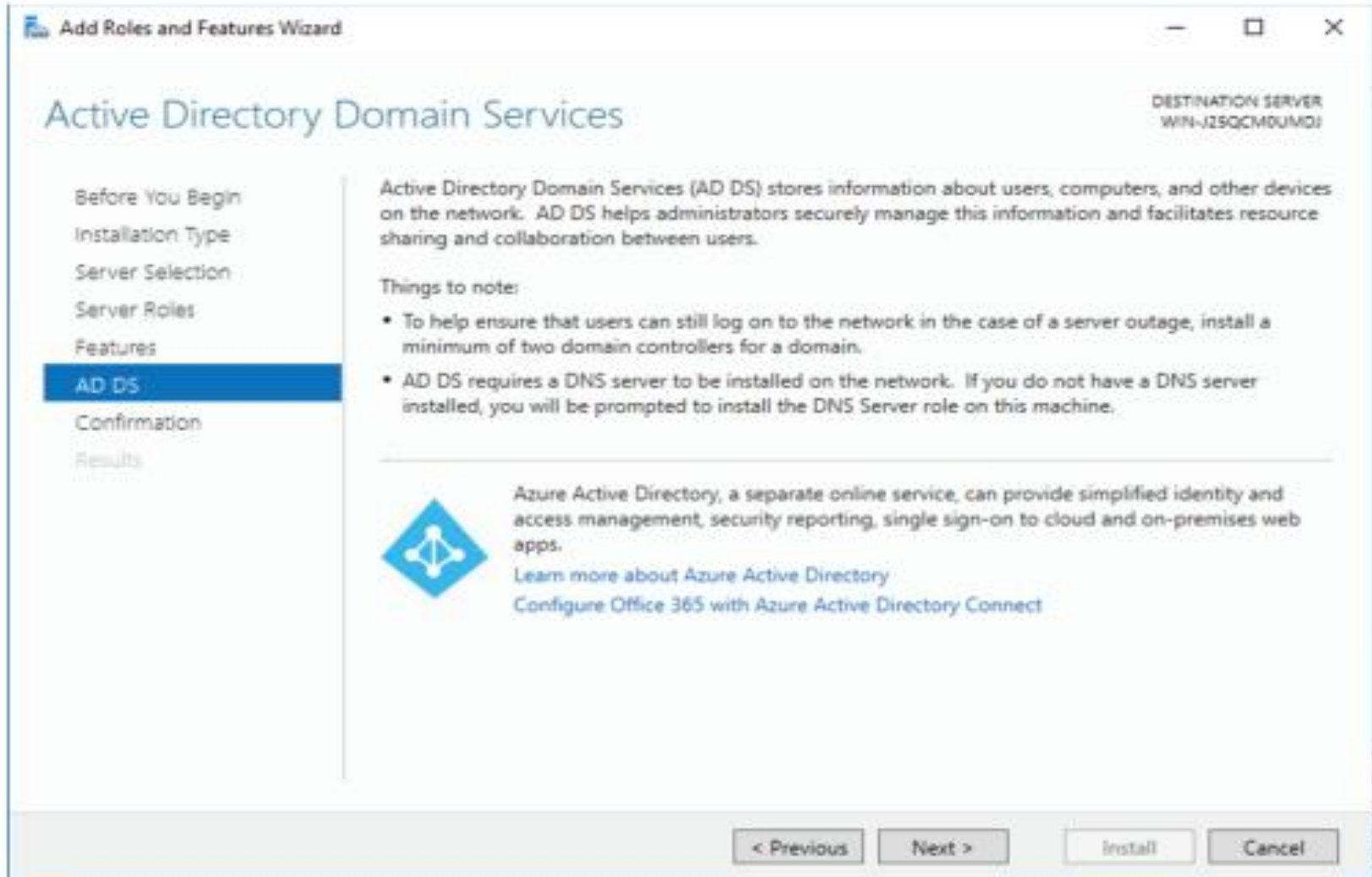
❑ Click Checkbox to select Active Directory Domain Services



❑ On the popup Window, just click Add Features.



- ❑ On the description window of Active Directory Domain Services, click Next.



❑ Click Install on the Confirmation window

The screenshot shows the 'Add Roles and Features Wizard' window. The title bar reads 'Add Roles and Features Wizard'. The main heading is 'Confirm installation selections'. In the top right corner, it says 'DESTINATION SERVER WIN-J25QCM0UMDJ'. On the left, a navigation pane lists the steps: 'Before You Begin', 'Installation Type', 'Server Selection', 'Server Roles', 'Features', 'AD DS', 'Confirmation' (which is highlighted in blue), and 'Results'. The main area contains the text: 'To install the following roles, role services, or features on selected server, click Install.' Below this is a checkbox labeled 'Restart the destination server automatically if required'. A paragraph follows: 'Optional features (such as administration tools) might be displayed on this page because they have been selected automatically. If you do not want to install these optional features, click Previous to clear their check boxes.' A list of features to be installed is shown in a box: 'Active Directory Domain Services', 'Group Policy Management', 'Remote Server Administration Tools', 'Role Administration Tools', 'AD DS and AD LDS Tools', 'Active Directory module for Windows PowerShell', 'AD DS Tools', 'Active Directory Administrative Center', and 'AD DS Snap-Ins and Command-Line Tools'. At the bottom of the main area, there are links for 'Export configuration settings' and 'Specify an alternate source path'. The bottom of the window has four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

Add Roles and Features Wizard

DESTINATION SERVER
WIN-J25QCM0UMDJ

Confirm installation selections

Before You Begin
Installation Type
Server Selection
Server Roles
Features
AD DS
Confirmation
Results

To install the following roles, role services, or features on selected server, click Install.

☐ Restart the destination server automatically if required

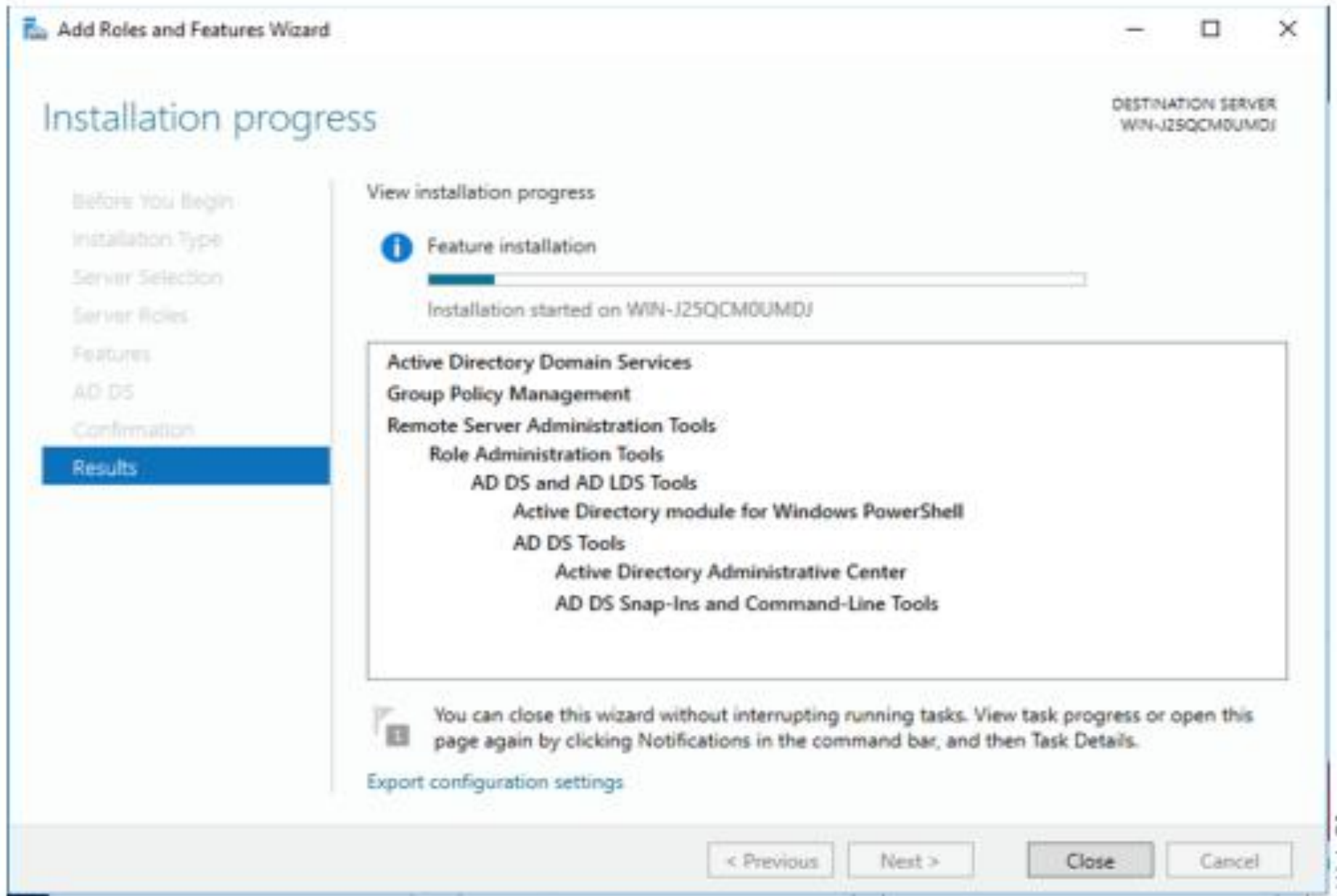
Optional features (such as administration tools) might be displayed on this page because they have been selected automatically. If you do not want to install these optional features, click Previous to clear their check boxes.

- Active Directory Domain Services
- Group Policy Management
- Remote Server Administration Tools
 - Role Administration Tools
 - AD DS and AD LDS Tools
 - Active Directory module for Windows PowerShell
 - AD DS Tools
 - Active Directory Administrative Center
 - AD DS Snap-Ins and Command-Line Tools

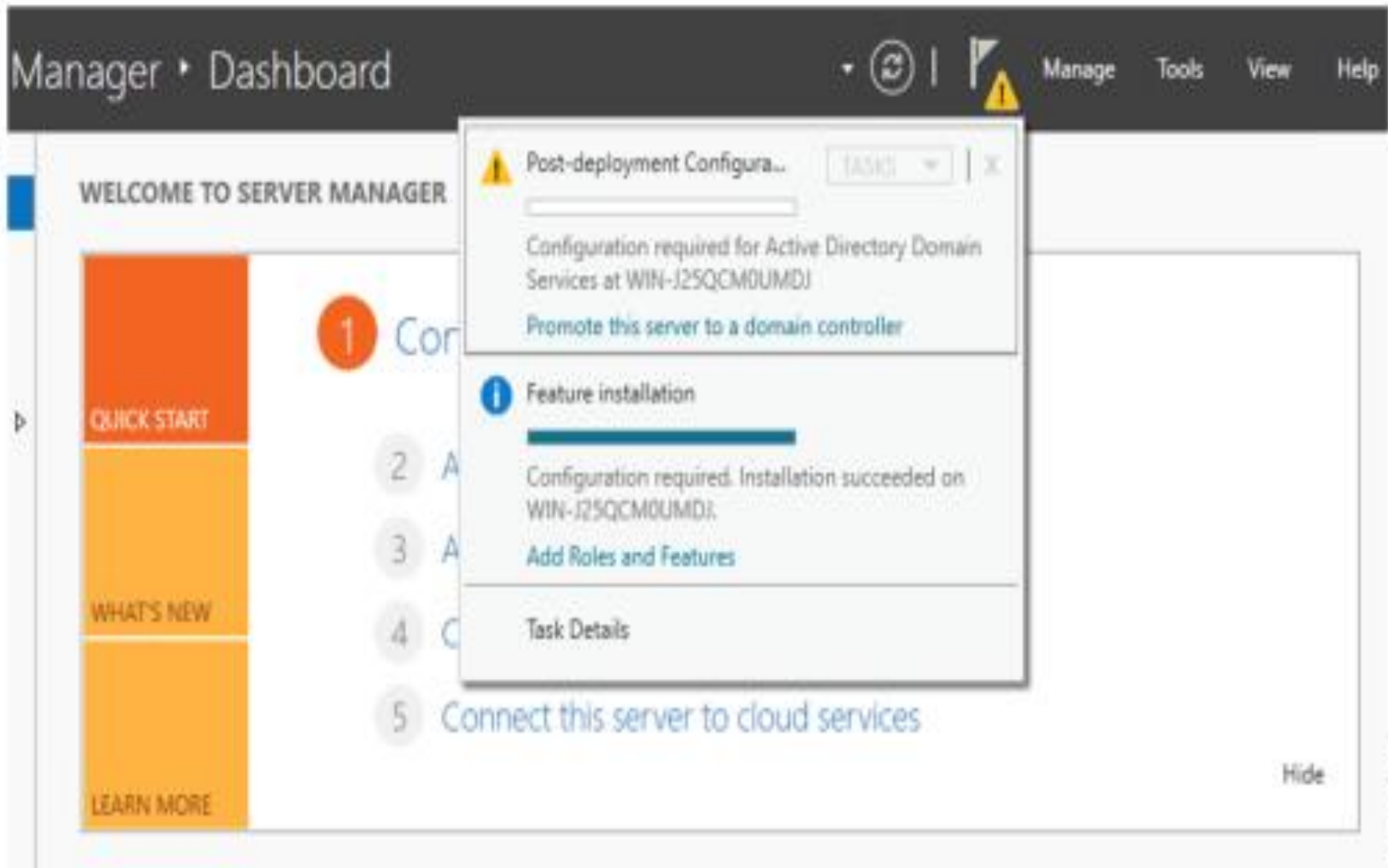
Export configuration settings
Specify an alternate source path

< Previous Next > Install Cancel

Installation process begins.



- ❑ After installing AD DS Role, you can promote this Server to a Domain Controller



- ❑ Select Add a new forest and give the Root domain name, ad.gpfs.net. Click Next

The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window. The title bar includes the Windows logo and the text 'Active Directory Domain Services Configuration Wizard'. The window has standard minimize, maximize, and close buttons. The main content area is titled 'Deployment Configuration'. On the right side, it says 'TARGET SERVER WIN-J25QCM0UMDJ'. On the left, there is a navigation pane with the following items: 'Deployment Configuration' (highlighted in blue), 'Domain Controller Options', 'Additional Options', 'Paths', 'Review Options', 'Prerequisites Check', 'Installation', and 'Results'. The main area contains the following text: 'Select the deployment operation' followed by three radio button options: 'Add a domain controller to an existing domain', 'Add a new domain to an existing forest', and 'Add a new forest' (which is selected). Below this, it says 'Specify the domain information for this operation' followed by a label 'Root domain name:' and a text box containing 'ad.gpfs.net'. At the bottom of the main area, there is a link 'More about deployment configurations'. At the very bottom of the window, there are four buttons: '< Previous', 'Next >' (highlighted in blue), 'Install', and 'Cancel'.

Active Directory Domain Services Configuration Wizard

Deployment Configuration

TARGET SERVER
WIN-J25QCM0UMDJ

Deployment Configuration

Domain Controller Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select the deployment operation

☐ Add a domain controller to an existing domain

☐ Add a new domain to an existing forest

☒ Add a new forest

Specify the domain information for this operation

Root domain name:

More about deployment configurations

< Previous Next > Install Cancel

❑ Enter the Directory Services Restore Mode password

The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window. The title bar includes standard Windows window controls. The main heading is 'Domain Controller Options'. On the right, it identifies the 'TARGET SERVER' as 'WIN-J25QCM0UMDJ'. A left-hand navigation pane lists the following steps: 'Deployment Configuration', 'Domain Controller Options' (which is highlighted with a blue background), 'DNS Options', 'Additional Options', 'Paths', 'Review Options', 'Prerequisites Check', 'Installation', and 'Results'. The main content area is divided into sections. The first section, 'Select functional level of the new forest and root domain', contains two dropdown menus: 'Forest functional level' and 'Domain functional level', both currently set to 'Windows Server 2016'. The second section, 'Specify domain controller capabilities', features three checkboxes: 'Domain Name System (DNS) server' (checked), 'Global Catalog (GC)' (checked), and 'Read only domain controller (RODC)' (unchecked). The third section, 'Type the Directory Services Restore Mode (DSRM) password', contains two password input fields labeled 'Password:' and 'Confirm password:', both of which are filled with ten dots. At the bottom of the main content area, there is a link that says 'More about domain controller options'. The bottom of the window features a navigation bar with four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'. The 'Next >' button is highlighted with a blue border.

Active Directory Domain Services Configuration Wizard

TARGET SERVER
WIN-J25QCM0UMDJ

Domain Controller Options

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Select functional level of the new forest and root domain

Forest functional level: Windows Server 2016

Domain functional level: Windows Server 2016

Specify domain controller capabilities

☒ Domain Name System (DNS) server
☒ Global Catalog (GC)
☐ Read only domain controller (RODC)

Type the Directory Services Restore Mode (DSRM) password

Password:

Confirm password:

[More about domain controller options](#)

< Previous Next > Install Cancel

❑ Ignore the warning message

Active Directory Domain Services Configuration Wizard

TARGET SERVER
WIN-J25QCM0UMDJ

DNS Options

A delegation for this DNS server cannot be created because the authoritative parent zone cannot be found... [Show more](#)

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Specify DNS delegation options

☐ Create DNS delegation

[More about DNS delegation](#)

< Previous Next > Install Cancel

- ❑ Use the default NetBIOS domain name and click Next

The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window. The title bar includes the Windows logo and the text 'Active Directory Domain Services Configuration Wizard'. The window has standard minimize, maximize, and close buttons. The main heading is 'Additional Options'. In the top right corner, it says 'TARGET SERVER WIN-J25QCM0UMDJ'. A yellow warning banner at the top states: 'The default value for the NetBIOS domain name is already being used, one alternative has been suggested. Show more X'. On the left is a navigation pane with the following items: 'Deployment Configuration', 'Domain Controller Options', 'DNS Options', 'Additional Options' (which is highlighted with a blue background), 'Paths', 'Review Options', 'Prerequisites Check', 'Installation', and 'Results'. The main content area contains the text 'Verify the NetBIOS name assigned to the domain and change it if necessary' and 'The NetBIOS domain name:' followed by a text box containing 'AD0'. At the bottom left of the main area is a link 'More about additional options'. The bottom of the window features four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

Active Directory Domain Services Configuration Wizard

Additional Options

TARGET SERVER
WIN-J25QCM0UMDJ

⚠ The default value for the NetBIOS domain name is already being used, one alternative has been suggested. [Show more](#) X

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Verify the NetBIOS name assigned to the domain and change it if necessary

The NetBIOS domain name:

[More about additional options](#)

< Previous Next > Install Cancel

- ❑ Use the default paths and click Next.

The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window. The title bar includes the Windows logo and the text 'Active Directory Domain Services Configuration Wizard'. The window has standard minimize, maximize, and close buttons. On the left is a navigation pane with the following items: 'Deployment Configuration', 'Domain Controller Options', 'DNS Options', 'Additional Options', 'Paths' (highlighted with a blue background), 'Review Options', 'Prerequisites Check', 'Installation', and 'Results'. The main area is titled 'Paths' in large blue font. In the top right corner, it says 'TARGET SERVER: WIN-J25QCM0UMDJ'. Below the title, a subtitle reads 'Specify the location of the AD DS database, log files, and SYSVOL'. There are three input fields with browse buttons (three dots in a box) to their right: 'Database folder:' with 'C:\Windows\NTDS', 'Log files folder:' with 'C:\Windows\NTDS', and 'SYSVOL folder:' with 'C:\Windows\SYSVOL'. At the bottom left of the main area is a link that says 'More about Active Directory paths'. The bottom of the window contains four buttons: '< Previous', 'Next >' (highlighted with a blue border), 'Install', and 'Cancel'.

Active Directory Domain Services Configuration Wizard

Paths

TARGET SERVER
WIN-J25QCM0UMDJ

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Specify the location of the AD DS database, log files, and SYSVOL

Database folder: C:\Windows\NTDS

Log files folder: C:\Windows\NTDS

SYSVOL folder: C:\Windows\SYSVOL

[More about Active Directory paths](#)

< Previous Next > Install Cancel

❑ Review and click Next if no errors

The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window. The title bar includes the Windows logo and the text 'Active Directory Domain Services Configuration Wizard'. In the top right corner are standard window controls (minimize, maximize, close). The main content area is titled 'Review Options' in a large blue font. On the left side, there is a vertical list of steps: 'Deployment Configuration', 'Domain Controller Options', 'DNS Options', 'Additional Options', 'Paths', 'Review Options' (highlighted with a blue background), 'Prerequisites Check', 'Installation', and 'Results'. The main area displays a summary of the configuration choices. It starts with 'Review your selections:' followed by a list of options: 'Configure this server as the first Active Directory domain controller in a new forest.', 'The new domain name is "ad.gpfs.net". This is also the name of the new forest.', 'The NetBIOS name of the domain: AD0', 'Forest Functional Level: Windows Server 2016', 'Domain Functional Level: Windows Server 2016', and 'Additional Options:'. Under 'Additional Options', there are three items: 'Global catalog: Yes', 'DNS Server: Yes', and 'Create DNS Delegation: No'. At the bottom of the main area, there is a note: 'These settings can be exported to a Windows PowerShell script to automate additional installations' with a 'View script' button to its right. Below this is a link: 'More about installation options'. At the very bottom of the window is a navigation bar with four buttons: '< Previous', 'Next >' (highlighted with a blue border), 'Install', and 'Cancel'.

Active Directory Domain Services Configuration Wizard

TARGET SERVER
WIN-J25QCM0UMDJ

Review Options

- Deployment Configuration
- Domain Controller Options
- DNS Options
- Additional Options
- Paths
- Review Options**
- Prerequisites Check
- Installation
- Results

Review your selections:

- Configure this server as the first Active Directory domain controller in a new forest.
- The new domain name is "ad.gpfs.net". This is also the name of the new forest.
- The NetBIOS name of the domain: AD0
- Forest Functional Level: Windows Server 2016
- Domain Functional Level: Windows Server 2016
- Additional Options:
 - Global catalog: Yes
 - DNS Server: Yes
 - Create DNS Delegation: No

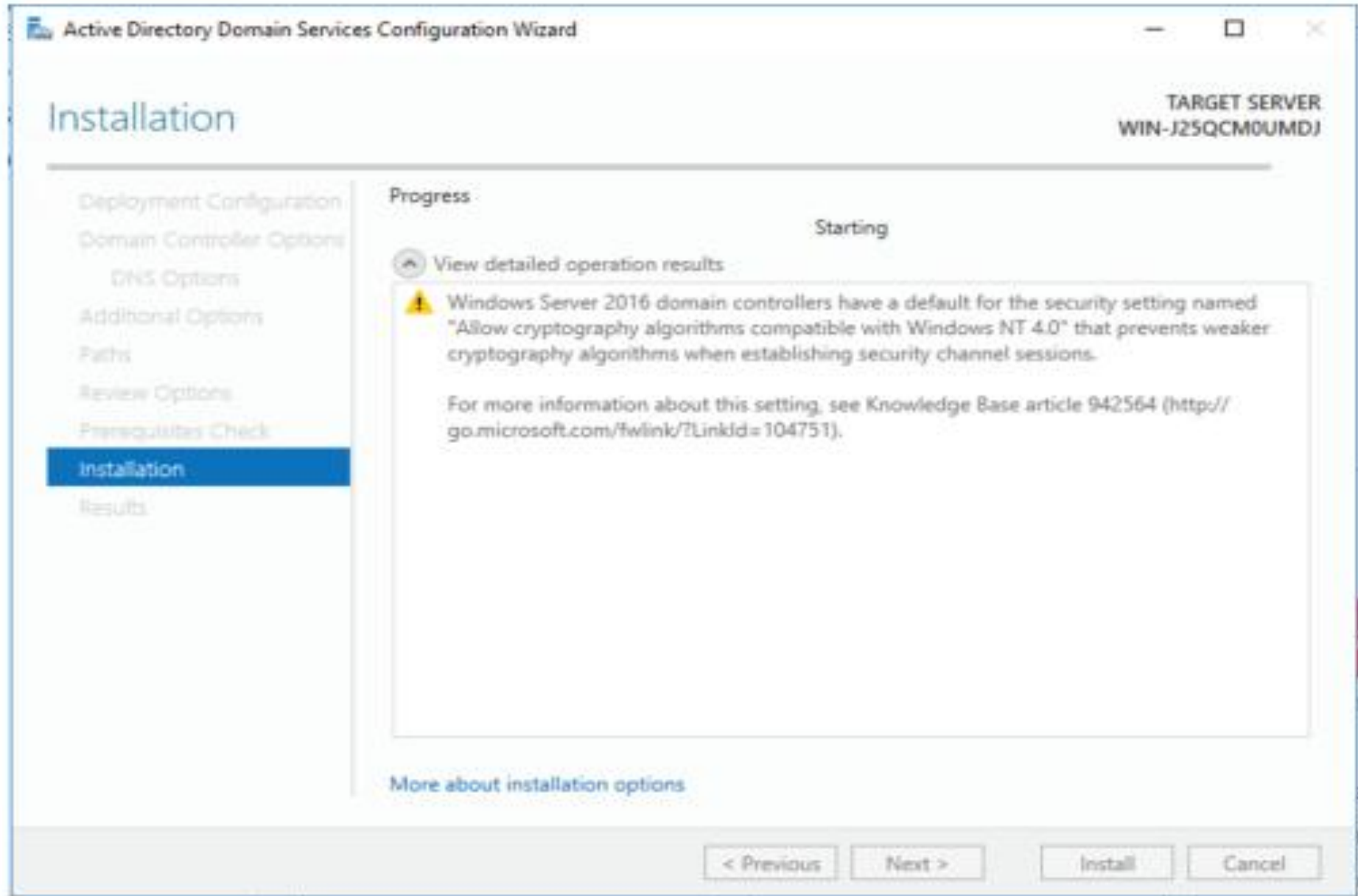
These settings can be exported to a Windows PowerShell script to automate additional installations

[View script](#)

[More about installation options](#)

< Previous **Next >** Install Cancel

- ❑ Click Install and wait for the installation to finish.



The Domain Controller is now set up.

Features of server operating systems

- ❑ **Multi-User Support:** Server OS can handle multiple users at once, managing resources efficiently to ensure smooth operation during peak usage times.
- ❑ **Security and Access Control:** Robust security features allow administrators to set permissions and control access to sensitive data, protecting against unauthorized access.
- ❑ **Resource Management:** These operating systems optimize hardware resources (CPU, memory, storage) for maximum efficiency and performance.

- ❑ **Scalability and Reliability:** Server OS are designed to scale easily with increased workloads and provide features like failover and backup to ensure continuous operation.
- ❑ **Networking Capabilities:** They support various networking protocols, enabling communication between servers and clients for functions such as file sharing and email hosting.

Types of server operating systems

- ❑ Windows server: Windows server is known for its user-friendly interface and integration with other Microsoft products.
- ❑ It supports enterprise-level management and supports Active Directory, group policies, and centralized domain management.
- ❑ Versions of Windows Server include Windows Server 2016, 2019, 2022, and others, each offering enhanced features over its predecessor.

- ❑ **Linux Server:** An open-source operating system that is highly regarded for its stability and security.
- ❑ Linux is widely used in web servers and enterprise applications due to its flexibility and cost-effectiveness.
- ❑ There are many Linux distributions tailored for servers, including *Ubuntu Server, CentOS/RHEL (Red Hat Enterprise Linux), Debian, Fedora Server, and SUSE Linux Enterprise Server*.
- ❑ Linux is highly customizable, often controlled via command-line interface (CLI), although graphical user interfaces (GUIs) are available for some distributions.

That's all for today

